

Q1

1) For P to be valid:

- every element must be positive, and less than or equal to 1.
- the sum of each row must be 1

Case 1: all entries positive ✓
all entries ≤ 1 ✓
all rows sum to 1 ✓

- row 1 sum = $0.5 + 0.5 = 1$
- row 2 sum = $0.4 + 0.6 = 1$

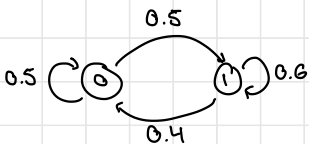
Case 2: all entries positive ✓
all entries ≤ 1 ✓
all rows sum to 1 ✓

- row 1 sum = $0 + 1 = 1$
- row 2 sum = $1 + 0 = 1$

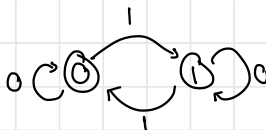
Case 3: all entries positive ✓
all entries ≤ 1 ✓
all rows sum to 1 ✓

- row 1 sum = $0.6 + 0.3 + 0.1 = 1$
- row 2 sum = $0.2 + 0.5 + 0.3 = 1$
- row 3 sum = $0.1 + 0.4 + 0.5 = 1$

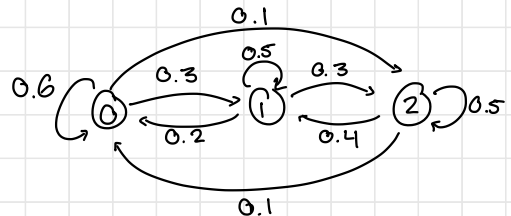
2) Case 1



Case 2



Case 3



$$3) \text{ Case 1: } \pi_1 = \pi_0 P_1 = [0.8 \ 0.2] \begin{bmatrix} 0.5 & 0.5 \\ 0.4 & 0.6 \end{bmatrix} = [0.48 \ 0.52]$$

$$\pi_2 = \pi_1 P_1 = [0.48 \ 0.52] \begin{bmatrix} 0.5 & 0.5 \\ 0.4 & 0.6 \end{bmatrix} = [0.448 \ 0.552]$$

$$\pi_3 = \pi_2 P_1 = [0.448 \ 0.552] \begin{bmatrix} 0.5 & 0.5 \\ 0.4 & 0.6 \end{bmatrix} = [0.4448 \ 0.5552]$$

$$\text{Case 2: } \pi_1 = \pi_0 P_2 = [1 \ 0] \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = [0 \ 1]$$

$$\pi_2 = \pi_1 P_2 = [0 \ 1] \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = [1 \ 0]$$

$$\pi_3 = \pi_2 P_2 = [1 \ 0] \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = [0 \ 1]$$

$$\text{Case 3: } \pi_1 = \pi_0 P_3 = [0.2 \ 0.5 \ 0.3] P_3 = [0.25 \ 0.43 \ 0.32]$$

$$\pi_2 = \pi_1 P_3 = [0.25 \ 0.43 \ 0.32] P_3 = [0.268 \ 0.418 \ 0.314]$$

$$\pi_3 = \pi_2 P_3 = [0.268 \ 0.418 \ 0.314] P_3 = [0.2758 \ 0.415 \ 0.3092]$$

4) See my code submission

5) Case 1: $\pi = [\pi_1 \ \pi_2]$, solve $\{ \pi P_1 = \pi, \sum_i \pi_i = 1 \}$ for π_1 and π_2

solved using mathematica: $\pi = [\pi_1 \ \pi_2] = [0.444 \ 0.556]$

```
In[16]:= Solve[{{\pi1, \pi2}.{{0.5, 0.5}, {0.4, 0.6}} == {\pi1, \pi2}, 1 == \pi1 + \pi2}, {\pi1, \pi2}]
```

```
Out[16]:= {{\pi1 -> 0.444444, \pi2 -> 0.555556}}
```

Case 2: same thing, $\pi = [\pi_1 \ \pi_2] = [\frac{1}{2} \ \frac{1}{2}]$

```
In[16]:= Solve[{{\pi1, \pi2, \pi3}.{{0.6, 0.3, 0.1}, {0.2, 0.5, 0.3}, {0.1, 0.4, 0.5}} == {\pi1, \pi2, \pi3}, 1 == \pi1 + \pi2 + \pi3}, {\pi1, \pi2, \pi3}]
```

```
Out[16]:= {{\pi1 -> 0.282609, \pi2 -> 0.413043, \pi3 -> 0.304348}}
```

Case 3: same again, $\pi = [\pi_1 \ \pi_2 \ \pi_3] = [0.283 \ 0.413 \ 0.304]$

```
In[16]:= Solve[{{\pi1, \pi2, \pi3}.{{0.6, 0.3, 0.1}, {0.2, 0.5, 0.3}, {0.1, 0.4, 0.5}} == {\pi1, \pi2, \pi3}, 1 == \pi1 + \pi2 + \pi3}, {\pi1, \pi2, \pi3}]
```

```
Out[16]:= {{\pi1 -> 0.282609, \pi2 -> 0.413043, \pi3 -> 0.304348}}
```

6) A chain is irreducible if all states in the state space are accessible from one another

A state is aperiodic if $P_{ii} > 0$

A chain is aperiodic if all states are aperiodic.

Case 1: can access all states from all other states: irreducible ✓

$$\left. \begin{array}{l} P_{11} > 0 \\ P_{22} > 0 \end{array} \right\} \text{aperiodic}$$

Case 2: can access all states from all other states: irreducible ✓

$$\left. \begin{array}{l} P_{11} = 0 \\ P_{22} = 0 \end{array} \right\} \text{periodic}$$

Case 3: can access all states from all other states: irreducible ✓

$$\left. \begin{array}{l} P_{11} > 0 \\ P_{22} > 0 \\ P_{33} > 0 \end{array} \right\} \text{aperiodic}$$

7) A Markov chain converges if:

- it is irreducible
- it is aperiodic

This means that a stationary distribution always exists, but if the above are satisfied then the chain will converge towards it.

So case 1 and 3 converge, but 2 does not.

Q2

1) transmission delay $D_{\text{tran}} = \frac{L}{R} = \frac{100 \times 8}{500 \times 10^6} = 1.6 \mu\text{s}$

2) $D_{\text{prop1}} = \frac{10}{2 \times 10^8} = 0.05 \mu\text{s}$

3) $D_{\text{prop2}} = \frac{15}{2 \times 10^8} = 0.075 \mu\text{s}$

4) a) Memory switching

• Only one packet can be transferred through memory at a time.

• If all 4 packets arrive at the same time, one may need to wait for the other 4-1 packets.

$$\text{max input queueing delay} = (N-1)D_{\text{prop}} = 3 \times 1.6 = 4.8 \mu\text{s}$$

b) Bus switching

• Same as memory, only one packet can use the bus at a time

$$\text{max input queueing delay} = (N-1)D_{\text{prop}} = 3 \times 1.6 = 4.8 \mu\text{s}$$

c) Crossbar switching

• Packets destined for different ports can be transferred at the same time.

• This is the case in this scenario, so

$$\text{max input queueing delay} = 0$$

5) a/b) Memory switching / Bus switching

$$D_{\text{total}} = D_{\text{trans}} + D_{\text{prop1}} + D_{\text{queue}} + D_{\text{trans}} + D_{\text{prop2}}$$

$$= 1.6 + 0.05 + 4.8 + 1.6 + 0.075 = 8.125 \mu\text{s}$$

c) Crossbar switching

$$D_{\text{total}} = D_{\text{trans}} + D_{\text{prop1}} + \cancel{D_{\text{queue}}} + D_{\text{trans}} + D_{\text{prop2}}$$

$$= 1.6 + 0.05 + 1.6 + 0.075 = 3.325 \mu\text{s}$$

6) End-to-end delay is the same for Memory/Bus, but lower for Crossbar.

This is because in Memory/Bus switching, there is a shared resource that the packets contend for, but in Crossbar switching there are multiple parallel paths, so multiple packets can move to their respective output ports at the same time.

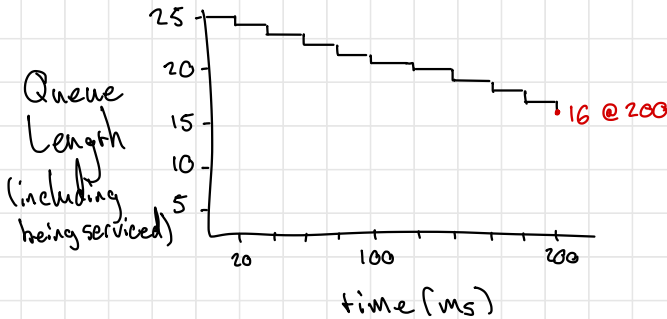
Q3

- a) 1, 2, 2, 2, 3 $\rightarrow$ repeat until buffer cleared.
- b) Same pattern, but when it looks for a class 3 packet to service and there isn't one, it skips it.

1, 2, 2, 2 $\rightarrow$ repeat

Q4

1)



2) 10 packets have been transmitted

3) 15 packets, including currently in transmission

4) a) The ACK's arrival rate is the same as the service rate, so the queue will stay at the same size (persistent)

b) Remains constant at 15

5) • Assuming 15 packets in queue @ 200 ms, front one just starting to be serviced.

• First ACK arrives @ 200 ms

• I will assume the new packet gets placed AHEAD of the ACK-generated packet at the back of the queue

So the new arriving packet has to wait $15 \times 20 = 300$ ms

6) This is simply the total time spent in the system.

$$\text{Queueing delay} + \text{transmission delay} = 300 + 20 = 320$$

7) This is an example of bufferbloat, because the router maintains a persistent queue, so we have a long queueing delay.

Throughput and delay are related, but separate. With bufferbloat, we end up with high latency simply because the queue is large, regardless of the high throughput / low packet loss.

Buffer bloat means we sacrifice latency for throughput.

8) If we limit the standing queue to 5, the delay is:

$$D_q = 5 \times 20 = 100 \text{ ms}$$

This is much shorter than in Q5.

It will reduce latency, which is especially useful for delay-sensitive applications.

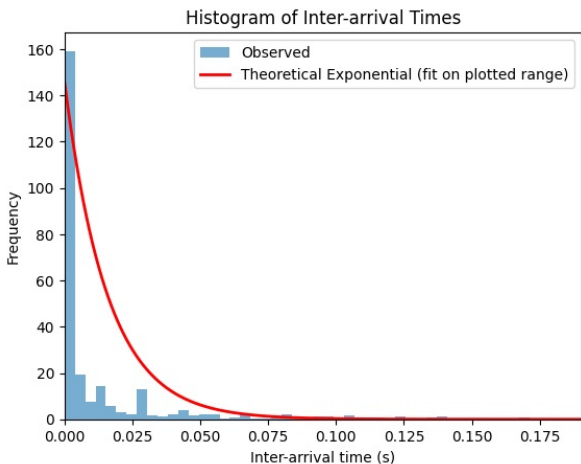
Q5

Here is a screenshot of a section of my wireshark capture.
The full capture CSV can be found in my submission.
All code is also in my submission.

No.	Time	Source	Destination	Protocol	Length	Info
24795	26.767167	10.0.0.216	10.0.0.138	TCP	54	62501 → 53 [ACK] Seq=34 Ack=279 Win=65200 Len=0
24796	26.768868	13.107.226.31	10.0.0.216	TCP	66	443 → 62502 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1440 SACK_PERM WS=12
24797	26.768909	10.0.0.216	13.107.226.31	TCP	54	62502 → 443 [ACK] Seq=1 Ack=1 Win=66048 Len=0
24798	26.769202	10.0.0.216	13.107.226.31	TLSv1.3	1776	Client Hello (SHA=c.clarity.ms)
24799	26.771096	142.250.195.234	10.0.0.216	UDP	68	443 → 53034 Len=26
24800	26.772859	13.107.226.31	10.0.0.216	TCP	60	443 → 62502 [ACK] Seq=1 Ack=1441 Win=66008 Len=0
24801	26.772919	13.107.226.31	10.0.0.216	TCP	60	443 → 62502 [ACK] Seq=1 Ack=1723 Win=71600 Len=0
24802	26.773177	13.107.226.31	10.0.0.216	TLSv1.3	153	Hello Retry Request, Change Cipher Spec
24803	26.773314	10.0.0.216	13.107.226.31	TLSv1.3	590	Change Cipher Spec, Client Hello (SHA=c.clarity.ms)
24804	26.777455	13.107.226.31	10.0.0.216	TLSv1.3	1514	Server Hello, Application Data
24805	26.777674	10.0.0.216	142.250.195.174	QUIC	74	Protected Payload (KPB), DCID=f4a5d486152462b6
24806	26.778373	10.0.0.216	172.66.0.33	TCP	54	62254 → 443 [ACK] Seq=3286 Ack=168792 Win=64512 Len=0
24807	26.778682	13.107.226.31	10.0.0.216	TCP	1514	443 → 62502 [ACK] Seq=1500 Ack=2259 Win=74240 Len=1460 [TCP segment of a reassembled PDU]
24808	26.778699	10.0.0.216	13.107.226.31	TCP	54	62502 → 443 [ACK] Seq=2259 Ack=3820 Win=66048 Len=0
24809	26.779689	13.107.226.31	10.0.0.216	TCP	1230	443 → 62502 [PSH, ACK] Seq=3820 Ack=2259 Win=74240 Len=1176 [TCP segment of a reassembled PDU]
24810	26.780917	13.107.226.31	10.0.0.216	TCP	1514	443 → 62502 [ACK] Seq=4196 Ack=2259 Win=74240 Len=1460 [TCP segment of a reassembled PDU]
24811	26.780935	10.0.0.216	13.107.226.31	TCP	54	62502 → 443 [ACK] Seq=2259 Ack=5656 Win=66048 Len=0
24812	26.782149	13.107.226.31	10.0.0.216	TLSv1.3	1514	Application Data
24813	26.782465	13.107.226.31	10.0.0.216	TLSv1.3	373	Application Data, Application Data
24814	26.782481	10.0.0.216	13.107.226.31	TCP	54	62502 → 443 [ACK] Seq=2259 Ack=7435 Win=66048 Len=0
24815	26.783508	10.0.0.216	13.107.226.31	TLSv1.3	128	Application Data
24816	26.783616	10.0.0.216	13.107.226.31	TLSv1.3	246	Application Data
24817	26.783701	3.121.166.250	10.0.0.216	TCP	1514	443 → 62478 [ACK] Seq=4613 Ack=2433 Win=50688 Len=1460 [TCP segment of a reassembled PDU]
24818	26.783717	10.0.0.216	13.107.226.31	TLSv1.3	552	Application Data
24819	26.784926	3.121.166.250	10.0.0.216	TCP	1514	443 → 62478 [ACK] Seq=6073 Ack=2433 Win=50688 Len=1460 [TCP segment of a reassembled PDU]

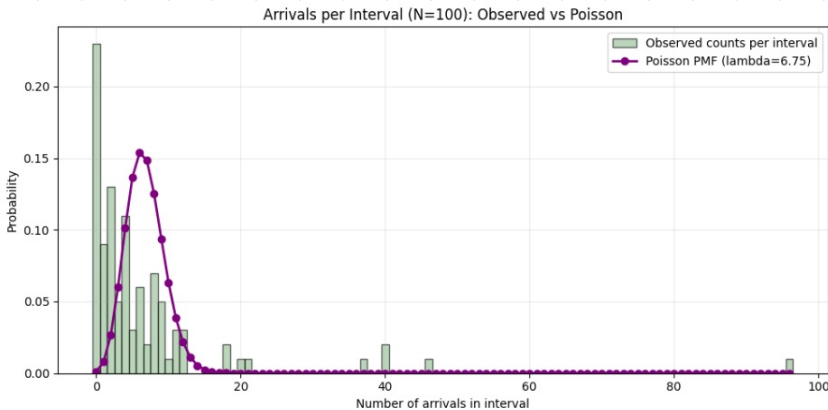
Frame 1: 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF{0439C7B-FCB6-41C9-B11A-CBC0B1DEADB5}, 0
 Ethernet II, Src: MicroStarInt_81:18:94 (2c:f0:5d:81:18:94), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
 Address Resolution Protocol (request)

Here I have computed the interarrival times, and used their mean to plot the theoretical exponential distribution to compare.



In general the packets received are exponentially distributed.
However it does not exactly match the theoretical. This is because the theoretical plot is based on the true mean, which is pulled upward by a few very large interarrival times

And here I have divided into equal time intervals and counted the arrivals in each interval, then calculated the mean arrivals per interval and used that to plot the theoretical Poisson distribution.



The observed traffic does not align very well with the theoretical Poisson distribution.

However, it does look like it might align with a Poisson PMF with a smaller value for λ , which I would have if I excluded the outliers

Similar to the previous plot, the few very large interarrival times + the bursty nature of many arrivals means that our plots do not align too well.

This is where we see the non idealities of reality start to diverge from the theory we learn.